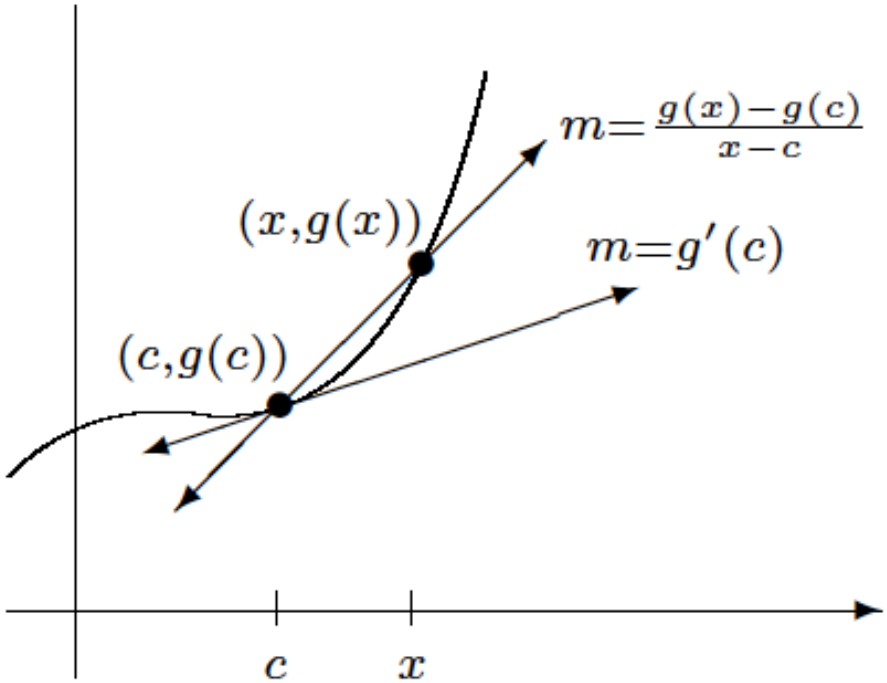
Readings

Simnon and Blume: 3.1 to 3.4 and 4.1 to 4.2

Derivative of function from R to R

The derivative of f at c is the slope of the graph of f at c.

It captures the 'rate of change in output' through the difference quotient $\frac{(f(x)-f(c))}{(x-c)}$.



Definition

f is **differentiable** at c if for all $\{x_n\}_{n=1}^{\infty}$ converging to c, the sequence of difference quotient $\left\{\frac{f(x_n)-f(c)}{x_n-c}\right\}_{n=1}^{\infty}$ converge to the same limit. The limit is known as the **derivative** of f at c.

equivalently, denoting $h_n = x_n - c$

f is **differentiable** at c if for all $\{h_n\}_{n=1}^{\infty}$ converging to 0, the sequence of difference quotient $\left\{\frac{f(c+h_n)-f(c)}{h_n}\right\}_{n=1}^{\infty}$ converges to the same limit. The limit is known as the **derivative** of f at c.

Notation

The limit of $\{f(x_n)\}_{n=1}^{\infty}$ when $\{x_n\}_{n=1}^{\infty}$ converges to c is denoted by

$$\lim_{x \to c} f(x)$$

The limit of $\left\{\frac{f(x_n)-f(c)}{x_n-c}\right\}_{n=1}^{\infty}$ when $\{x_n\}_{n=1}^{\infty}$ converges to c is denoted by

$$f'(c) = \lim_{x \to c}\left[\frac{f(x)-f(c)}{x-c}\right]$$

equivalently using h_n

$$f'(c) = \lim_{h \to 0}\left[\frac{f(c+h)-f(c)}{h}\right]$$

Examples

1. $f(x) = x^2$

$$f'(c) = \lim_{x \to c} \frac{(x^2 - c^2)}{(x - c)} = \lim_{x \to c} (x + c) = 2c$$

f is differentiable at all points on $\mathcal{R}$. Derivative of f at c is $2c$.

2. $f(x) = |x|$.

The difference quotient sequence $\left\{ \frac{|x|-0}{x-0} \right\}_{n=1}^{\infty}$ have different limits depending on whether we take sequences of positive or negative numbers (converging to 0). Thus f is not differentiable at 0. However f is differentiable at all other points on $\mathcal{R}$.

Differentiablity and Continuity

Result: If f is differentiable at c then f is continuous at c. The converse is not true.

Algebra of Derivatives

Let f and g be differentiable at c. Then

- $(f+g)'(c) = f'(c) + g'(c)$ and $(f-g)'(c) = f'(c) - g'(c)$
- $(fg)'(c) = f'(c)g(c) + g'(c)f(c)$
- If $g(c) \neq 0$ then $\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - g'(c)f(c)}{(g(c))^2}$

Chain Rule

If f is differentiable at c and g is differentiable at $f(c)$ then $(g \circ f)'(c) = g'(f(c))f'(c)$

Reinterpreting Derivative: Linear Approximation

What is a **linear approximation** of $f(y)$ when y is 'close' to c?

- A linear function is of the form $g(t) = \alpha + \beta t$.
- $g(c)$ should be equal to $f(c)$.
- $g(y)$ should be 'close' to $f(y)$ for all y 'close' to c. That is

$$\lim_{y \to c} \frac{[f(y) - g(y)]}{y - c} = 0$$

Reinterpreting Derivative: Linear Approximation

The first two conditions imply that $g(t) = f(c) + \beta(t - c)$

The second condition then implies

$$\lim_{y \to c} \frac{[f(y) - f(c) - \beta(y - c)]}{y - c} = 0$$

What is β?

We now approach the issue from the other side to show $\beta = f'(c)$ (when f is differentiable at c)

Reinterpreting Derivative: Linear Approximation

STEP 1: For sequences $\{x_n\}_{n=1}^{\infty}$ converging to c, the difference quotient sequence $\left\{\frac{f(x_n)-f(c)}{x_n-c}\right\}_{n=1}^{\infty}$ converges to $f'(c)$.

STEP 2: Using algebraic properties of converging sequences, we can say $\left\{\frac{f(x_n)-f(c)}{x_n-c} - f'(c)\right\}_{n=1}^{\infty}$ converges to 0.

$$\lim_{x\to c} \frac{[f(x) - f(c) - f'(c)(x-c)]}{x-c} = 0$$

Therefore $\beta = f'(c)$ would work for linear approximation

Reinterpreting Derivative: Linear Approximation

$[f(c) + f'(c)h]$is the linear approximation function. This is the tangent line of f at c.

Error in approximation:

Define

$$e(h) = [f(c+h) - f(c) - f'(c)h]$$

From STEP 2:

$$\lim_{h \to 0} \frac{e(h)}{h} = 0$$

Proof: Differentiability and Continuity

Result: If f is differentiable at c then f is continuous at c. The converse is not true.

Proof:

$$
\begin{aligned}
&\lim_{x\to c}[f(x)-f(c)] \\
=&\lim_{x\to c}\left[\frac{(f(x)-f(c))}{(x-c)}(x-c)\right] \\
=&\left[\lim_{x\to c}\frac{(f(x)-f(c))}{(x-c)}\right]\left[\lim_{x\to c}(x-c)\right] \\
=&f'(c)\cdot 0 \\
=&0
\end{aligned}
$$

Hence, $\lim_{x\to c} f(x) = f(c)$

We have already seen that $f(x) = |x|$ is not differentiable at 0. Check that it is continuous at 0.

Proof: Algebra of derivatives

Let f and g be differentiable at c.
Result 1 $(f+g)'(c) = f'(c) + g'(c)$

Proof:

$$\begin{aligned}(f+g)'(c) &= \lim_{x\to c} \frac{[f(x)+g(x)]-[f(c)+g(c)]}{x-c} \\ &= \lim_{x\to c} \frac{[f(x)-f(c)]}{x-c} + \lim_{x\to c} \frac{[g(x)-g(c)]}{x-c} \\ &= f'(c) + g'(c)\end{aligned}$$

Result 2: Suppose $f(c) \neq 0$ and $h(x) = \frac{1}{f(x)}$. Then $h'(c) = -\frac{f'(c)}{f(c)^2}$.

Proof:

$$\begin{aligned}h'(c) &= \lim_{x\to c} \frac{h(x)-h(c)}{x-c} \\ &= \lim_{x\to c} \left[-\frac{f(x)-f(c)}{(x-c)f(x)f(c)}\right] \\ &= \lim_{x\to c} \left[-\frac{f(x)-f(c)}{(x-c)}\right] \lim_{x\to c} \left[\frac{1}{f(x)f(c)}\right] \\ &= -f'(c)\frac{1}{f(c)^2} \quad [\text{Since } f \text{ is differentiable at } c, f \text{ is also continuous at } c]\end{aligned}$$

Proof: Chain Rule

Chain rule: If f is differentiable at c and g is differentiable at $f(c)$ then $(g \circ f)'(c) = g'(f(c))f'(c)$

Proof: From the discussion of best linear approximation, we know that

$$g(y) = g(f(c)) + g'(f(c))(y - f(c)) + e(y - f(c))$$

where $\lim_{y \to c} \frac{e(y-f(c))}{y-f(c)} = 0$

Choose $y = f(c+h)$ and dividing by $h \, (\neq 0)$

$$\frac{g(f(c+h)) - g(f(c))}{h} = g'(f(c)) \frac{[f(c+h) - f(c)]}{h} + \frac{e(f(c+h) - f(c))}{h}$$

Taking a sequence of h converging to 0, we have

$$(g \circ f)'(c) = g'(f(c))f'(c) + \lim_{h \to 0} \frac{e(f(c+h) - f(c))}{h}$$

Since f is also continuous at c (it is differentiable at c), $\lim_{h \to 0} f(c+h) = f(c)$.

$$\lim_{h \to 0} \frac{e(f(c+h) - f(c))}{h} = \left[\lim_{h \to 0} \frac{e(f(c+h) - f(c))}{f(c+h) - f(c)} \right] \left[\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \right] = 0 \cdot f'(c) = 0$$